Intro To Backend, What's API And RESTFull API

Assignment

Problem Statement:

You are tasked with designing and documenting the RESTful API backend for an **Online Bookstore System**. This assignment evaluates your technical understanding of backend architecture, request/response lifecycles, RESTful design principles, and structured JSON data exchange.

Task 1: Backend Architecture & Request Lifecycle

Suppose a user clicks "Add to Cart" on the bookstore's web frontend.

1. Describe the complete journey of this request from the browser to the backend server and back to the client. Explicitly explain the roles of:
    - DNS lookup and IP addresses
    - Standard HTTP/HTTPS protocols and ports
    - Server, Application/Business Logic Layer, and Database
2. Detail the specific responsibilities of the Application/Business Logic layer during this interaction (e.g., authentication checks, inventory availability, business logic rules).

Task 2: RESTful Endpoint Design

Design a set of RESTful endpoints to manage the bookstore's inventory (Books) and user purchases (Orders). Fill in the blank cells in the table below, specifying the appropriate HTTP Method, URL Path, Safe/Idempotent properties, and a brief description for each requirement.

Design Guidelines:

- Use resource-based, lowercase, plural nouns in path parameters.
- Apply standard HTTP verbs (GET, POST, PUT, PATCH, DELETE) appropriately.
- Include API versioning in all endpoint paths (e.g., /v1/...).

Requirement / Operation	HTTP Method	Endpoint (URL Path)	Safe? (Yes/No)	Idempotent ? (Yes/No)	Brief Description
Fetch list of all books					
Fetch details of a specific book by ID					
Add a new book to the inventory					
Update price of an existing book					
Delete a book record from inventory					
Place a new book order					
Retrieve all orders belonging to a specific					

Requirement / Operation	HTTP Method	Endpoint (URL Path)	Safe? (Yes/No)	Idempotent ? (Yes/No)	Brief Description
user					

Task 3: Data Exchange & JSON Payloads

For the **Place a new book order** operation designed in Task 2, construct the HTTP request and response structures for both success and failure scenarios:

1. **Successful Order Creation (201 Created):**
   - ○ Specify the required request headers (e.g., Content-Type, Authorization). ○ Write the request JSON body containing relevant order fields (e.g., book ID, quantity, shipping address).
   - ○ Write the expected successful JSON response body returned by the backend server.
2. **Error Handling Scenario:**
   - ○ Describe a specific failure scenario (e.g., requested quantity exceeds stock, or invalid data field).
   - ○ Specify the appropriate HTTP Status Code (e.g., 400, 409, or 422).
   - ○ Construct a structured JSON error response body that includes error codes, human-readable messages, and targeted field indicators.

Deliverables:

● Submit a written document (or Markdown file) containing the detailed answers for Task 1, the completed design table for Task 2, and the constructed JSON payloads for Task 3. ● If you implemented code or mock servers, upload the code to GitHub and share the Pull Request (PR) link and branch name.
● Send status updates over email upon completing each milestone of the assignment.